

of these shapes so that the animator can create asymmetry in the facial expressions. Shapes for the upper part of the face include blinks and brow positions for emotions such as worry, anger, and surprise. Again, it is a good idea to create left and right versions of these shapes to give the animator more control.

6.6.2 Progressive Morphs

Morphs typically work by moving the vertices of the object from one position to another along a straight line. This default motion works great for many applications such as facial animation, but can be problematic when morphing an object that needs to move along an arc, such as the joints of a finger or eyelids moving across the round surface of the eyeball. Progressive morphing addresses this problem by allowing a morph to pass through multiple targets, giving you the ability to create in between shapes.

6.7 Wiring parameters

3ds Max provides tools that allow you to connect the action of one parameter to another. This is called wiring and is useful in character rigging as well as other areas. Wiring allows you to create control panels for characters and areas such as facial animation. It also allows animation along one part of a character to affect others.

Follow the following steps to try it out yourself

1. Right-click on the object and select Wire Parameters.
2. Select the parameter to wire. In the case of a cylinder its radius is selected.
3. Drag the link to the control object and select the parameter used as the control. In this case, the box's Z position is used to control the radius.
4. This brings up the Parameter Wiring window, where the parameters and control direction can be set. In addition, expressions can be added.
5. The wiring is complete. When the box is moved along Z, the radius of the cylinder changes to match.
6. Parameter wiring is used to create control panels, such as this one used to control facial animation.
7. 3ds Max has manipulators, which allow you to create floating sliders in a viewport. These can also be used with parameter wiring to create control panels. 